

IZPITNA VPRAŠANJA

I. KINEMATIKA TOČKE

1. Definicija $\vec{v}$. hitrosti, pospeška; geometrijski pomen

Gibanje boke:

$$\vec{r}(t) = x(t)\vec{e}_x + y(t)\vec{e}_y + z(t)\vec{e}_z$$

$$\vec{v} = \frac{d\vec{r}}{dt} = \dot{\vec{r}} = \dot{x}\vec{e}_x + \dot{y}\vec{e}_y + \dot{z}\vec{e}_z \rightarrow \text{HITROST (sprememba položaja po času)}$$

$$|\vec{v}| = \sqrt{\dot{x}^2 + \dot{y}^2 + \dot{z}^2} \rightarrow \text{BRZINA}$$

Geom. pomen:

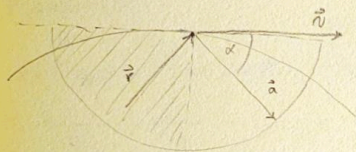


$\vec{v}$ tangenta na tir

$$\vec{a} = \frac{d\vec{v}}{dt} = \ddot{\vec{r}} = \ddot{x}\vec{e}_x + \ddot{y}\vec{e}_y + \ddot{z}\vec{e}_z \rightarrow \text{POSPESEK (sprememba hitrosti po času)}$$

$$\vec{a} = \ddot{x}\vec{e}_x + \ddot{y}\vec{e}_y + \ddot{z}\vec{e}_z$$

Geom. pomen:



$\vec{a}$ kaže smer zavijanja +, če:

$$\alpha \in (0, \frac{\pi}{2}) \rightarrow \text{pospešeno} \Leftrightarrow \vec{a} \cdot \vec{v} > 0$$

$$\alpha = \frac{\pi}{2} \rightarrow \text{enakomerno} \Leftrightarrow \vec{a} \cdot \vec{v} = 0$$

$$\alpha \in (\frac{\pi}{2}, \pi) \rightarrow \text{zavirajoče} \Leftrightarrow \vec{a} \cdot \vec{v} < 0$$

Preučimo gibanje:

$$x = x(t)$$

$$x = x(t)$$

$$v = \dot{x}(t)$$

$$a = \ddot{x}(t)$$

Enakomerno gibanje: $\dot{x} = v_0 \Rightarrow x = v_0 t + c$

$$x_0 = v_0 t_0 + c$$

$$\Rightarrow x = v_0(t - t_0) + x_0$$

Enakomerno pospešeno gibanje: $a = a_0 = \ddot{x}$

$$v = a_0(t - t_0) + v_0 \text{ in}$$

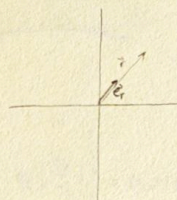
$$\dot{x} = v \Rightarrow \dot{x}(t - t_0) = v_0$$

$$x = \frac{1}{2} a_0(t - t_0)^2 + v_0(t - t_0) + x_0$$

2. Vektor hitrosti, pospeška v polarnem KS.

$$\vec{r}(t) = r(t) \cdot \vec{e}_r(t)$$

↑ RADIALNI BAZIS $\vec{v}$.



$$\vec{v} = \dot{\vec{r}} = \dot{r}\vec{e}_r + r \frac{d\vec{e}_r}{dt} = \dot{r}\vec{e}_r + r\dot{\phi}\vec{e}_\phi$$

$$\frac{d\vec{e}_r}{dt} = \frac{d\vec{e}_r}{d\phi} \frac{d\phi}{dt} = \frac{d}{d\phi}(\cos\phi\vec{e}_x + \sin\phi\vec{e}_y) \cdot \dot{\phi}$$

$$= (-\sin\phi\vec{e}_x + \cos\phi\vec{e}_y) \cdot \dot{\phi}$$

$$= \vec{e}_\phi \cdot \dot{\phi}$$

$$\vec{a} = \dot{\vec{v}} = \frac{\partial}{\partial t}(\dot{r}\vec{e}_r + r\dot{\phi}\vec{e}_\phi) = \underbrace{(\ddot{r} - r\dot{\phi}^2)}_{\text{RADIALNI } a} \vec{e}_r + \underbrace{(r\ddot{\phi} + 2\dot{r}\dot{\phi})}_{\text{OBODNI } a} \vec{e}_\phi$$

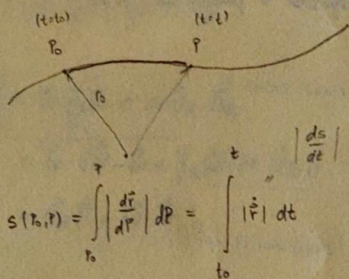
IZPRAVILA:

$$\vec{a} = \dot{\vec{v}} = \frac{\partial}{\partial t}(\dot{r}\vec{e}_r + r\dot{\phi}\vec{e}_\phi) = \frac{\partial}{\partial t}(\dot{r}\vec{e}_r) + \frac{\partial}{\partial t}(r\dot{\phi}\vec{e}_\phi) =$$

$$= \ddot{r}\vec{e}_r + \dot{r}\dot{\phi}\vec{e}_\phi + (\dot{r}\dot{\phi} + r\ddot{\phi})\vec{e}_\phi - r\dot{\phi}\vec{e}_r$$

$$= (\ddot{r} - r\dot{\phi}^2)\vec{e}_r + (r\ddot{\phi} + 2\dot{r}\dot{\phi})\vec{e}_\phi$$

3. Kinematika gibanja po krivulji



s - NARAVNI PARAMETER

$$\vec{v} = \dot{s} \vec{e}_t \quad \text{NORMALNA KOMPONENTA}$$

$$\vec{a} = \ddot{s} \vec{e}_t + \dot{s} \dot{\vec{e}}_t \quad \vec{a} \in \text{prijetena-ravnina (K)}$$

TANGENCIJNA KOMPONENTA

$$\kappa = \frac{|\vec{v} \times \vec{a}|}{|\vec{v}|^3} = \frac{|\dot{x}\dot{y} - \dot{y}\dot{x}|}{|\dot{x} + \dot{y}|^{3/2}} \rightarrow \text{FLEKSIJSKA UKRIVLJENOST}$$

$$\vec{e}_b = \vec{e}_t \times \vec{e}_n$$

$$\frac{d\vec{e}_b}{ds} = -\tau \vec{e}_n \quad \left(\vec{B} = -\tau \vec{N} \right)$$

↑ TORZIJSKA UKRIVLJENOST

II DINAMIKA TÖKE

1. Primeri sil in njihovi potenciali

Newtoni zakoni:

INZ: KS je inercialen (natočno ni zmagajoči sil) $\Leftrightarrow$ točno se giblje
pomočito s konstantno brzino

II NZ: V KS velja: $F = ma$

$$\text{III NZ: } \vec{F}_{12} = -\vec{F}_{21}$$

A. KONSTANTNA SILA

$$\vec{F} = \vec{F}_0$$

$$\vec{F} = m\vec{g}$$

$$U = -\vec{F} \cdot \vec{r} = -F_x x - F_y y - F_z z$$

$$\frac{\partial U}{\partial x} = -F_x$$

$$\text{grad } U = -\vec{F}$$

B. GRAVITACIJSKA SILA

$$\vec{F} = -\frac{\gamma m_1 m_2}{r^2} \vec{r}$$

$$U = -\frac{\gamma m_1 m_2}{r}$$

$$\vec{F} = -\text{grad } U \quad \frac{x}{r}$$

$$\frac{\partial U}{\partial x} = -\gamma m_1 m_2 \left(-\frac{1}{r^2} \right) \frac{\partial r}{\partial x}$$

C. ELEKTRIČNA SILA

$$\vec{F}_e = -\frac{e_1 e_2}{|P_2 - P_1|^2} \cdot \frac{P_2 - P_1}{|P_2 - P_1|}$$

$$U = \frac{e_1 e_2}{r}$$

D. ELEKTRO-MAGNETNA SILA

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$$

$\Rightarrow$ odvisno od $v \Rightarrow$ ni konzervativna

E. SILA VZMETA

$$\vec{F} = -kx$$

$$\vec{F} = -k(|\vec{r} - \vec{r}_0| - L) \cdot \frac{\vec{r} - \vec{r}_0}{|\vec{r} - \vec{r}_0|} \quad \text{! Samo za majhne razdalje}$$

$$U = \frac{1}{2} k (|\vec{r} - \vec{r}_0| - L)^2$$

F. SILA UPORA

$$\vec{F} = -f(\vec{v}) \cdot \frac{\vec{v}}{|\vec{v}|}$$

$$\text{LINEARN: } \vec{F} = -k\vec{v}$$

$$\text{KVADRATNI: } \vec{F} = -k|\vec{v}| \vec{v}$$

$$\text{KONSTANTNI: } f(|\vec{v}|) = f_0 \Rightarrow \text{TRESE}$$

$$f_0 = k|\vec{v}|$$

2. IZREK O DELU, ENERGIJI

$$A = \int_{P_1}^{P_2} \vec{F} \cdot d\vec{r} = \int_{t_1}^{t_2} \underbrace{\vec{F} \cdot \vec{v}}_{\text{moč}} dt$$

I (o delu): A enako različni kinetične E $\Rightarrow$ v KS vsja izred o delu.

$$\begin{aligned} \text{D: } A &= \int_{t_1}^{t_2} \vec{F} \cdot \vec{v} dt = \int_{t_1}^{t_2} m \vec{a} \cdot \vec{v} dt = m \int_{t_1}^{t_2} \vec{v} \cdot \frac{d\vec{v}}{dt} dt = m \int_{t_1}^{t_2} \frac{d}{dt} \left(\frac{1}{2} \vec{v}^2 \right) dt = \\ &= m \int_{t_1}^{t_2} \frac{d}{dt} \left(\frac{1}{2} \dot{r}^2 \right) dt = m \int_{t_1}^{t_2} \frac{d}{dt} \left(\frac{1}{2} v^2 \right) dt = m \cdot \frac{1}{2} v^2 \Big|_{t_1}^{t_2} \\ &= \frac{1}{2} m v_2^2 - \frac{1}{2} m v_1^2 = T_2 - T_1 \end{aligned}$$

Def: Sila je POTENCIJALNA, če obstoja skokana fun. U (potencial), da:

$$\vec{F} = -\text{grad } U$$

$$\text{Ij } \vec{F} \text{ oblike: } \vec{F} = - \left(\frac{\partial U}{\partial x} \vec{i} + \frac{\partial U}{\partial y} \vec{j} + \frac{\partial U}{\partial z} \vec{k} \right)$$

Def: $\vec{F}$ KONSERVATIVNA, če je potencialna in odvisna samo od položaja.

$$\vec{F} = -\text{grad } U; \quad U = U(\vec{r})$$

Ij $\vec{F}$ konservativna:

$$\begin{aligned} A &= \int_{t_1}^{t_2} \vec{F} \cdot \vec{v} dt = - \int_{t_1}^{t_2} \text{grad } U \cdot \vec{v} dt = - \int_{t_1}^{t_2} \text{grad } U \cdot \frac{d\vec{r}}{dt} dt \\ &= - \int_{t_1}^{t_2} \left(\frac{\partial U}{\partial x} \dot{x} + \frac{\partial U}{\partial y} \dot{y} + \frac{\partial U}{\partial z} \dot{z} \right) dt = - \int_{t_1}^{t_2} \frac{dU}{dt} dt = U_2 - U_1 \end{aligned}$$

$$\Rightarrow A = U_2 - U_1 = T_2 - T_1$$

$$\Rightarrow U_1 + T_1 = U_2 + T_2 = E_0$$

I (o E): Če je sila konservativna, je vsota kinetične in potencialne energije konstanta gibanja: $T + U = E_0$.

III PREMOŠČITNO GIBANJE

1. KVALITATIVNA OBRAVNAVA

PREMOŠČITNO GIBANJE = 3 KS, v katerem je gibanje po premici

$$\text{Če } f(x) = f \Rightarrow m\ddot{x} = f(t, x, \dot{x}) = f(x) \text{ in } U = - \int_{x_0}^x f(x) dx + U(x_0)$$

! če f zvezo

$$\text{Izrek o E: } \frac{1}{2} m \dot{x}^2 + U(x) = E_0$$

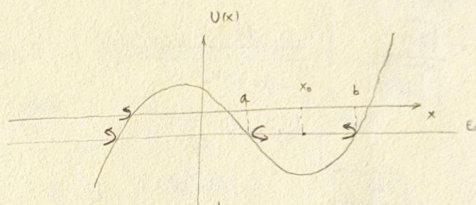
$$t = \text{sgn}(\dot{x}) \int_{x_0}^x \frac{dx}{\left(\frac{2}{m} (E_0 - U(x)) \right)} + t_0$$

$$\dot{x} = \pm \sqrt{\frac{2}{m} (E_0 - U(x))}$$

$$\pm \frac{dx}{\sqrt{\frac{2}{m} (E_0 - U(x))}} = dt$$

$$\Rightarrow \text{sgn}(\dot{x}) \cdot \int_{x_0}^x \frac{dx}{\sqrt{\frac{2}{m} (E_0 - U(x))}} = \int_{t_0}^t dt = t - t_0$$

$$\Rightarrow t = t(x) \text{ čas prelaza v točki}$$



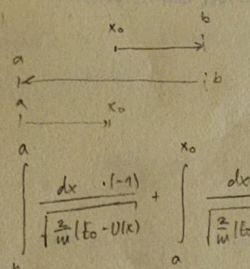
$$\text{Perioda gibanja: } T = \sqrt{2m} \int_a^b \frac{dx}{\sqrt{E_0 - U(x)}}$$

Minimum = STABILNA TOČKA

Maximum in preoj = NESTABILNA TOČKA

2. izračun periode, harmonična aproksimacija

Izvoris periode:



The diagram shows a potential well with energy levels E_0 and E_1 . The turning points are labeled a and b . The width of the well is x_0 .

$$T = \int_{x_0}^b \frac{dx}{\sqrt{\frac{2}{m}(E_0 - U(x))}} + \int_b^a \frac{dx \cdot (-1)}{\sqrt{\frac{2}{m}(E_0 - U(x))}} + \int_a^{x_0} \frac{dx}{\sqrt{\frac{2}{m}(E_0 - U(x))}}$$

$$= 2 \int_a^b \frac{dx}{\sqrt{\frac{2}{m}(E_0 - U(x))}} = \frac{2\sqrt{m}}{\sqrt{2}} \int_a^b \frac{dx}{\sqrt{E_0 - U(x)}} = \sqrt{2m} \int_a^b \frac{dx}{\sqrt{E_0 - U(x)}}$$

HARMONIČNA APROKSIMACIJA:

$$U(x) \approx U(x_0) + U'(x_0)(x - x_0) + \frac{1}{2}U''(x_0)(x - x_0)^2$$

$$T \approx \sqrt{2m} \int_{\tilde{a}}^{\tilde{b}} \frac{dx}{\sqrt{E_0 - (U(x_0) + U'(x_0)(x - x_0) + \frac{1}{2}U''(x_0)(x - x_0)^2)}}$$

$$= \frac{\sqrt{2m}}{\sqrt{\frac{1}{2}U''(x_0)}} \int_{\tilde{a}}^{\tilde{b}} \frac{dx}{\sqrt{-(x - x_0)^2 - \dots}}$$

$$T_0 = 2\pi \sqrt{\frac{m}{U''(x_0)}}$$

$$\int_a^b \frac{dx}{\sqrt{p_2(x)}} = \pi$$

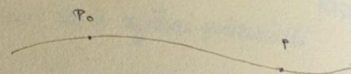
$$\begin{aligned} \rightarrow \deg(p_2) &= 2 \\ \rightarrow p_2''(x) &< 0 \quad (\text{vredni koef} < 0) \\ \rightarrow p_2(a) &= p_2(b) = 0 \end{aligned}$$

Libracijska:

$$T_1 = \pi \sqrt{\frac{m}{2}} \left(\frac{1}{\chi(a)} - \frac{1}{\chi(b)} \right);$$

$$E_0 - U(x) = (b-x)(x-a)\chi(x)$$

3. Gibanje po kugli;



$$S = S(t)$$

$$\vec{v} = \dot{S} \vec{e}_t$$

$$\vec{a} = \dot{S} \vec{e}_t + S \dot{\vec{e}}_t$$

$$m(\dot{S} \vec{e}_t + S \dot{\vec{e}}_t) = \vec{F} + \vec{S} \leftarrow \begin{matrix} \text{AKTIVNA SILA} & \text{SILA VEZI} \end{matrix}$$

$$\vec{e}_t + m\ddot{S} = \vec{F} \vec{e}_t + \dot{S} \dot{\vec{e}}_t$$

$$\vec{e}_n: m\dot{S}^2 = \vec{F} \cdot \vec{e}_n + \dot{S} \dot{\vec{e}}_n$$

$$\vec{e}_t: 0 = \vec{F} \cdot \vec{e}_t + \dot{S} \dot{\vec{e}}_t$$

3. enote, 2. neznanli

1) Gibanje po gladni kugli (brez kugle)

$$\dot{S} \cdot \vec{e}_t = 0$$

2) Gibanje po hrupni

$$|\dot{S} \cdot \vec{e}_t| = k \cdot \sqrt{(\dot{S} \cdot \vec{e}_n)^2 + (\dot{S} \cdot \vec{e}_t)^2}$$

IV GIBANJE V POLJU CENTRALNE SILE

1. Integralni gibanja

Def: Sila $\vec{F} = \vec{F}(P)$ je CENTRALNA, če $\exists$ točka O (CENTER SILE), da:

$$\vec{F} = F(|P-O|) \cdot \frac{P-O}{|P-O|} = F(r) \cdot \frac{\vec{r}}{r} = F(r) \cdot \vec{e}_r$$

VERIČNA KONČNINA: $\vec{L} = \vec{L}(P, O) = (P-O) \times m\vec{P} = \vec{r} \times m\vec{v}$

POL

$$\dot{\vec{L}} = \dot{\vec{r}} \times m\vec{v} + \vec{r} \times m\dot{\vec{v}} = \vec{r} \times \vec{F} = \vec{r} \times F(r)\vec{e}_r = \vec{0}$$

T: Konzervativna sila je centralna $\Leftrightarrow$ obstaja točka O , da $\vec{L}(P, O)$ konstanta gibanja

č. Fov. je vsaka konzervativna sila centralna in

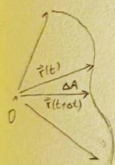
$$U = - \int_{r_0}^r F(r) dr + U_0$$

T: Gibanje v polju centralnih sil ima dve konstanti:

- L (vrednost količine)
- E (energija)

T: Gibanje po polju centralne sile je ravninsko

PROJEKCIJA HITROST



$$\Delta A = \frac{1}{2} \vec{r}(t) \times \vec{r}(t + \Delta t)$$

$$= \frac{1}{2} \vec{r}(t) \times (\vec{r}(t + \Delta t) - \vec{r}(t))$$

$$\frac{\Delta A}{\Delta t} = \frac{1}{2} \vec{r}(t) + \frac{1}{2} (\vec{r}(t + \Delta t) - \vec{r}(t))$$

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta A}{\Delta t} = \frac{1}{2} \vec{r}(t) + \vec{v}(t)$$

$$\vec{A} = \frac{1}{2} \vec{r} \times \vec{v} \quad \vec{L} = 2m\vec{A}$$

$$|\vec{r} \times \vec{v}| = C_0 = \text{DVOJNA PROJEKCIJA HITROST}$$

$$L = |\vec{L}| = m C_0$$

$$V \text{ polja: } C_0 = |\vec{r} \times \vec{v}| = r^2 \dot{\varphi} ; \dot{\varphi} > 0$$

2. Keplerjevi zakoni, Binetova formula enačba

IK1: Planeti se gibljejo okrog sonca po elipsi s soncem v goriščni elipse.

II K2: Radij $\vec{r}$ od sonca do planeta opiše v evoluti krivulji evolutno ploščino

III K2: Kvadrat periode je sorazmeren kubu večje polosi elipse.

$$T: C_0 = C_0 = \text{konst.} \Rightarrow A_0 = 0$$

$$D: A_0 = r^2 \ddot{\varphi} + 2\dot{r}\dot{\varphi}$$

$$r A_0 = r^3 \ddot{\varphi} + 2r\dot{r}\dot{\varphi} = (r^2 \dot{\varphi})' = C_0$$

BINETOVA ENAČBA / FORMULA:

$$A_r = -C_0^2 u^2 (1 + u'') ; u = \frac{1}{r}$$

V elipsi:

$$A_r = -\frac{C_0^2}{P} \cdot \frac{1}{r^2} ; P \text{ goriščni parameter}$$

$$r = \frac{P}{1 + E \cos \varphi}$$

izpovedi:

$$r = r(\varphi)$$

$$A_r = \ddot{r} - r\dot{\varphi}^2$$

$$\dot{r} = \frac{dr}{d\varphi} \cdot \dot{\varphi}$$

$$r^2 \dot{\varphi} = C_0$$

$$\ddot{r} = \frac{dr}{d\varphi} \cdot \frac{d\dot{\varphi}}{d\varphi}$$

$$u = \frac{1}{r}$$

$$u' = \frac{du}{d\varphi} = -\frac{1}{r^2} \cdot \frac{dr}{d\varphi}$$

$$\Rightarrow \dot{r} = -C_0 u'$$

$$\ddot{r} = \frac{d}{dt} (-C_0 u') = -C_0 u'' \cdot \dot{\varphi} = -C_0^2 \frac{1}{r^2} u'' = -C_0^2 u'^2 u''$$

$$\Rightarrow A_r = -C_0^2 u'^2 u'' \cdot r \cdot \frac{C_0^2}{r^4} = -C_0^2 u'^2 u'' - \frac{C_0^2}{r^3} = -C_0^2 u'^2 u'' - u^3 C_0^2$$

$$\Rightarrow A_r = -C_0^2 u^2 (u'' + u)$$

3. Redukcija na premoćno gibanje

Naj V tak, da $\vec{F} = -\text{grad } V$

$$\frac{1}{2} m v^2 + V(r) = E_0$$

$$\dot{\varphi} = \frac{L_0}{r^2}$$

$$\vec{v} = \dot{r} \vec{e}_r + r \dot{\varphi} \vec{e}_\varphi$$

$$v^2 = \vec{v} \cdot \vec{v} = \dot{r}^2 + (r \dot{\varphi})^2 = \dot{r}^2 + \frac{L_0^2}{r^2}$$

$$\frac{1}{2} m \left(\dot{r}^2 + \frac{L_0^2}{r^2} \right) + V(r) = E_0$$

$$\frac{1}{2} m \dot{r}^2 + \frac{m L_0^2}{2 r^2} + V(r) = E_0 \quad L = m L_0$$

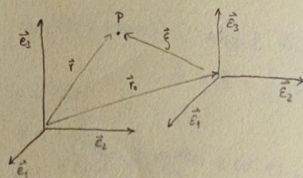
$$\frac{1}{2} m \dot{r}^2 + \frac{L^2}{2 m r^2} + V(r) = E_0$$

REDUKCIJA NA PREMOĆNO GIBANJE

$$U(r) = \frac{L^2}{2 m r^2} + V(r) \Rightarrow \text{EFEKTIVNI POTENCIJAL}$$

V. RELATIVNO GIBANJE

1. Vektor kotne hitrosti



$$\vec{r} = \vec{r}_0 + \vec{s}$$

$$\vec{s} = s_1 \vec{e}_1 + s_2 \vec{e}_2 + s_3 \vec{e}_3$$

$$\vec{v}_m = \dot{s}_1 \vec{e}_1 + \dot{s}_2 \vec{e}_2 + \dot{s}_3 \vec{e}_3$$

$$\vec{v} = \dot{r}_0 + \dot{\vec{s}}$$

$$= \dot{r}_0 + \dot{s}_1 \vec{e}_1 + \dot{s}_2 \vec{e}_2$$

$$\dot{\vec{e}}_i = a_{i1} \vec{e}_1 + a_{i2} \vec{e}_2 + a_{i3} \vec{e}_3$$

$$0 = \dot{\vec{e}}_i \cdot \vec{e}_i = a_{ii}$$

$$0 = \dot{\vec{e}}_i \cdot \vec{e}_j \Rightarrow \dot{\vec{e}}_i \cdot \vec{e}_j = -\vec{e}_i \cdot \dot{\vec{e}}_j$$

$$\begin{aligned} \dot{\vec{e}}_1 &= a_{12} \vec{e}_2 + a_{13} \vec{e}_3 \\ \dot{\vec{e}}_2 &= a_{21} \vec{e}_1 + a_{23} \vec{e}_3 \\ \dot{\vec{e}}_3 &= a_{31} \vec{e}_1 + a_{32} \vec{e}_2 \end{aligned} \Rightarrow \begin{aligned} \dot{\vec{e}}_1 \cdot \vec{e}_2 &= a_{12} \\ \dot{\vec{e}}_2 \cdot \vec{e}_1 &= a_{21} \end{aligned} \Rightarrow a_{12} = a_{21}$$

$$A = [a_{ij}] \text{ ANTISIM.}$$

$$\exists \vec{\omega}, \vec{\omega}: \dot{\vec{e}}_i = \vec{\omega} \times \vec{e}_i$$

$$\vec{\omega} = \omega_1 \vec{e}_1 + \omega_2 \vec{e}_2 + \omega_3 \vec{e}_3$$

$$\vec{\omega} = \vec{v} \text{ KOTNE HITOST}$$

$$T: (Q\dot{\vec{a}}) = Q\dot{\vec{a}} + \vec{\omega} \times Q\vec{a}$$

I $\vec{\omega}$ vektorje $R(\vec{e}, \varphi)$ okrog stalne osi $\vec{e}$ za kol φ $\vec{\omega} = \dot{\varphi} \vec{e}$

$$\vec{\omega} = \dot{\varphi} \vec{e} + \sin \varphi \vec{e} + (1 - \cos \varphi) \vec{e} \times \vec{e}$$

$\vec{\omega}$ KAŽE V SMERI TRENUTNE OSI ROTACIJE

2. Relativna kinost, pospeški

$$\vec{r} = \vec{r}_0 + \vec{\xi}; \quad \dot{\vec{\xi}} = \dot{\xi}_i \vec{e}_i$$

$$\vec{v}_{rel} = \dot{\xi}_i \vec{e}_i$$

$$\exists \vec{\omega}, \dot{\vec{e}}_i = \vec{\omega} \times \vec{e}_i$$

$$\vec{v} = \dot{\vec{r}} = \dot{\vec{r}}_0 + \dot{\xi}_i \vec{e}_i + \xi_i \dot{\vec{e}}_i$$

$$= \dot{\vec{r}}_0 + \dot{\xi}_i \vec{e}_i + \xi_i \vec{\omega} \times \vec{e}_i$$

$$= \dot{\vec{r}}_0 + \vec{\omega} \times \vec{\xi} + \vec{v}_{rel}$$

↑
TRANSLACIJSKA

↑
ROTACIJSKA

RELATIVNA

$$\vec{a} = \dot{\vec{v}} = \ddot{\vec{r}}_0 + \dot{\vec{\omega}} \times \vec{\xi} + \vec{\omega} \times \dot{\vec{\xi}} + \dot{\vec{v}}_{rel}$$

$$\dot{\vec{\xi}} = (\dot{\xi}_i \vec{e}_i)' = \dot{\xi}_i \dot{\vec{e}}_i + \vec{\omega} \times \vec{\xi} = \vec{\omega} \times \vec{\xi} + \vec{v}_{rel}$$

$$\dot{\vec{v}}_{rel} = (\dot{\xi}_i \vec{e}_i)' = \ddot{\xi}_i \vec{e}_i + \dot{\xi}_i \dot{\vec{e}}_i = \ddot{\xi}_i \vec{e}_i + \dot{\xi}_i \vec{\omega} \times \vec{e}_i$$

$$= \ddot{\xi}_i \vec{e}_i + \vec{\omega} \times \dot{\xi}_i \vec{e}_i$$

$$\vec{a}_{rel} = \ddot{\xi}_i \vec{e}_i$$

$$\vec{a} = \ddot{\vec{r}}_0 + \dot{\vec{\omega}} \times \vec{\xi} + \vec{\omega} \times (\vec{\omega} \times \vec{\xi}) + \vec{\omega} \times \vec{v}_{rel} + \vec{a}_{rel} + \vec{\omega} \times \vec{v}_{rel}$$

$$\vec{a} = \ddot{\vec{r}}_0 + \dot{\vec{\omega}} \times \vec{\xi} + \vec{\omega} \times (\vec{\omega} \times \vec{\xi}) + 2\vec{\omega} \times \vec{v}_{rel} + \vec{a}_{rel}$$

↑ ↑ ↑ ↑ ↑
INERCIJSKA EULERIJEVA CENTRIFUGALNA CORIOLISOVA RELATIVNA
POSP.

↓
SISTEMSKA

3. Newtonova enačba v RKS

$$m\vec{a} = \vec{F}$$

$$m\vec{a}_{rel} = \vec{F} - \underbrace{m\ddot{\vec{r}}_0}_{\text{INERCIJSKA}} - \underbrace{m\dot{\vec{\omega}} \times \vec{\xi}}_{\text{EULERJEVA}} - \underbrace{m\vec{\omega} \times (\vec{\omega} \times \vec{\xi})}_{\text{CENTRIFUGALNA}} - \underbrace{2m\vec{\omega} \times \vec{v}_{rel}}_{\text{CORIOLISOVA}}$$

$$m\vec{a}_{rel} = \vec{F}_{rel}$$

Če se izkloščijo njevala: $\ddot{\vec{r}}_0 = \vec{0}$

Če se rotira enolično: $\dot{\vec{\omega}} = \vec{0}$

VI SISTEM MATERIALNIH TOČEK

1. Masno središče

$$\vec{r}_K = \frac{1}{M} \sum_{i=1}^N m_i \vec{r}_i \quad ; \quad M = \sum_{i=1}^N m_i$$

$$\Rightarrow \ddot{\vec{r}}_K = \frac{1}{M} \sum_{i=1}^N m_i \ddot{\vec{r}}_i$$

$$\Rightarrow M \ddot{\vec{r}}_K = \vec{F} \Rightarrow \text{ENAKOST GIBANJA MASENA SREDIŠČA}$$

2. Izvedba o vrtilni količini

$$\vec{L} = \vec{L}(O) = \sum_{i=1}^N (\vec{r}_i - O) \times m_i \vec{v}_i$$

$\uparrow$
POL

$$\frac{d\vec{L}}{dt}(O) = \sum_{i=1}^N \vec{v}_i \times m_i \vec{v}_i + \sum_{i=1}^N (\vec{r}_i - O) \times m_i \vec{a}_i$$

$$= \sum_{i=1}^N \vec{v}_i \times m_i \vec{v}_i + \sum_{i=1}^N \vec{r}_i \times m_i \vec{a}_i$$

$$= \sum_{i=1}^N \vec{r}_i \times \vec{F}_i + \sum_{i=1}^N \vec{r}_i \times \vec{F}_i$$

$$= \vec{N}_n(O) + \vec{N}(O')$$

$\uparrow$
Rezultata vanjske
notranje sile

$\uparrow$
" " "
zunanje sile

Def: Če $\vec{F}_i = F_{ji} (\vec{r}_i - \vec{r}_j) \cdot \frac{\vec{r}_i - \vec{r}_j}{|\vec{r}_i - \vec{r}_j|}$, $\vec{F}$ CENTRALNA

T: Če so vse sile centralne $\Rightarrow N_n(O) = 0$

I (o vrtilni količini): Če so vse notranje sile centralne, je odvod vrtilne količine obuzg fiksnoga pola O' enak rezultatu vanjskih sil obuzg istoga pola.

$$\frac{d\vec{L}}{dt}(O') = \vec{N}(O')$$

3. Energijska točka

VII KINEMATIKA TOGEGA TELESA

1. Opis gibanja togega telesa

→ Translatorsko gibanje opisano kot gibanje masnega središča

→ Rotacijsko gibanje opisano kot rotacija okrog m^* .

Dinamične enačbe togega telesa:

$$1) \quad m \vec{a}_x = \vec{F}$$

$$2) \quad \frac{d\vec{L}(P^*)}{dt} = \vec{N}(P^*)$$

Tensorški produkt: $(\vec{a} \otimes \vec{b}) \vec{r} = (\vec{b} \cdot \vec{r}) \vec{a}$

Uvedemo: TELESNI KOORDINATNI SISTEM (TKS)
(RKS s središčem v P^*).

Gibanje togega telesa opisemo glede na TKS glede na AKS
(ali prostorski KS)

$$\vec{r}_x = \frac{1}{m} \sum_{i=1}^N \vec{r}_i m_i \rightarrow \frac{1}{m} \int_B \vec{r}_i \rho dV$$

$$m = \sum_{i=1}^N m_i \rightarrow \int_B \rho dV$$

$$\vec{L}(P_0) = \int_B (\vec{r} - \vec{P}_0) \times (\dot{\vec{r}} - \dot{\vec{P}}_0) dm \quad \left(\sum_{i=1}^N (\vec{r}_i - \vec{P}_0) \times m_i (\dot{\vec{r}}_i - \dot{\vec{P}}_0) \right)$$

$$= \int_B \vec{\xi} \times (\vec{\omega} \times \vec{\xi}) dm$$

$$= \int_B [\vec{\xi}, \vec{\omega}, \vec{\xi}] dm$$

$$= \int_B (\vec{\xi} \cdot \vec{\xi}) \vec{\omega} - (\vec{\omega} \cdot \vec{\xi}) \vec{\xi} dm$$

$$= \int_B (|\vec{\xi}|^2 \vec{\omega} - (\vec{\xi} \otimes \vec{\xi}) \vec{\omega}) dm$$

$$= \int_B (|\vec{\xi}|^2 \mathbf{I} - \vec{\xi} \otimes \vec{\xi}) dm \cdot \vec{\omega}$$

$$= \underline{\underline{J(P_0)}} \vec{\omega}$$

2. Vektorski tenzor

$\underline{J}(P_0)$ vektorski tenzor

$\underline{J}_e = \hat{e} \underline{J}(P_0) \hat{e} \leftarrow$ VETROVOSTNI MOMENT OKOLI OSI $\hat{e}$

$$\underline{J}(P_0) = \underbrace{\int_B (\hat{r})^2 \underline{I} \, dm}_{\text{SFERICAL DEL}} - \underbrace{\int_B \hat{r} \otimes \hat{r} \, dm}_{\text{DEVATOR}}$$

$$\underline{J}_n = \hat{e}_i \underline{J}(P_0) \hat{e}_i = \int_B (\hat{r}_1^2 + \hat{r}_2^2 + \hat{r}_3^2 - \hat{r}_1^2) \, dm$$

$$e_i (\hat{r} \otimes \hat{r}) e_i = (\hat{r}_i \hat{r}_i) (\hat{e}_i \hat{e}_i) = \hat{r}_i^2$$

$$J_{ii} = \int_B \hat{r}_i^2 + \hat{r}_k^2 \, dm; \quad k, j \neq i$$

$$\hat{e} \underline{J}(P_0) \hat{e} = (\hat{e}_i \hat{r}) (\hat{e}_j \hat{r}) = \hat{r}_1^2 + \hat{r}_2^2 + \hat{r}_3 \hat{r}_1 + \hat{r}_3^2$$

$$J_{ij} = \int_B \hat{r}_1^2 + \hat{r}_2^2 + \hat{r}_3^2 - \hat{r}_i^2 - \hat{r}_j^2 - \hat{r}_i \hat{r}_j - \hat{r}_j \hat{r}_i - \hat{r}_3^2 \, dm$$

$$= - \int_B \hat{r}_i \hat{r}_j \, dm$$

$$J_{ij} = J_{ji}$$

T. Naj bo Σ ravnina zrcalne simetrije telesa B, $P_0 \in \Sigma$,

$\hat{e}$ normala Σ

$\Rightarrow \hat{e}$ v glavi smeri $\underline{J}(P_0)$

T. Naj ima B dve ravninski simetriji Σ_1 in Σ_2 , potem je

$\Sigma_1 \cap \Sigma_2$ glavna smer $\underline{J}(P_0)$ za $P_0 \in \Sigma_1 \cap \Sigma_2$

I. STEINER:

$$\underline{J}(P_0) = \underline{J}(P_A) + m |P_0 - P_A|^2 \underline{I} - m (P_0 - P_A) \otimes (P_0 - P_A)$$

VIII DINAMIKA TOGEGA TELESIA

1. Eulerjeva dinamična enačba

1. PRINCIP O GIBANJU MASNEGA SREDIŠČA

$$m \vec{a}_x = \vec{F}$$

2. PRINCIP O VRTLNI KOLIČINI

$$\frac{d\underline{L}(P_A)}{dt} = \underline{N}(P_A)$$

3.

$$\underline{N}(P_A) + (P_A - P_0) \times \vec{F} \quad \text{posledo simetrično}$$

$$\mathcal{F} = \{(P_i, F_i); i \in [n]\}$$

$\vec{R}(\mathcal{F}) =$ REZULTANTA SISTEMA SIL

$$\vec{N}(\mathcal{F}, O) = \sum_{i=1}^n (P_i - O) \times \vec{F}_i$$

Def: Dva sistema sil $\mathcal{F}$ in $\mathcal{F}'$ sta EKUIPOLENTNA, če:

$$\vec{R}(\mathcal{F}) = \vec{R}(\mathcal{F}')$$

$$\vec{N}(\mathcal{F}, O) = \vec{N}(\mathcal{F}', O)$$

T. Dva ekvipolentna sistema imata enake dinamične učinke na togo telo.

PRINCIP O POLOŽENI SILE:

Prijemalnice neke sile je lahko bpt. bpt. na premici v smeri sile

VOLUMENSKO / POVRŠINSKE / LINIJSKE SILE

$$\vec{F} \text{ je VOLUMENSKA } \vec{F} = \int_B \vec{f} \, dm$$

$$\vec{f} = \text{SPECIFIČNA GOSTOTA SILE}$$

II.

$$d\vec{F} = \vec{g} \, dm$$

$$\vec{F} = \int_B \vec{g} \, dm$$

$$\vec{F} = m \vec{g}$$

EULERJEVA DINAMIČNA ENAČBA:

$$\frac{d\underline{L}(P)}{dt} = \underline{J}(P) \vec{\omega} + \vec{\omega} \times \underline{J}(P) \vec{\omega}$$

I. Naj P_0 uinje. Potem ujo:

$$\frac{d\underline{L}(P_0)}{dt} = \underline{N}(P_0)$$